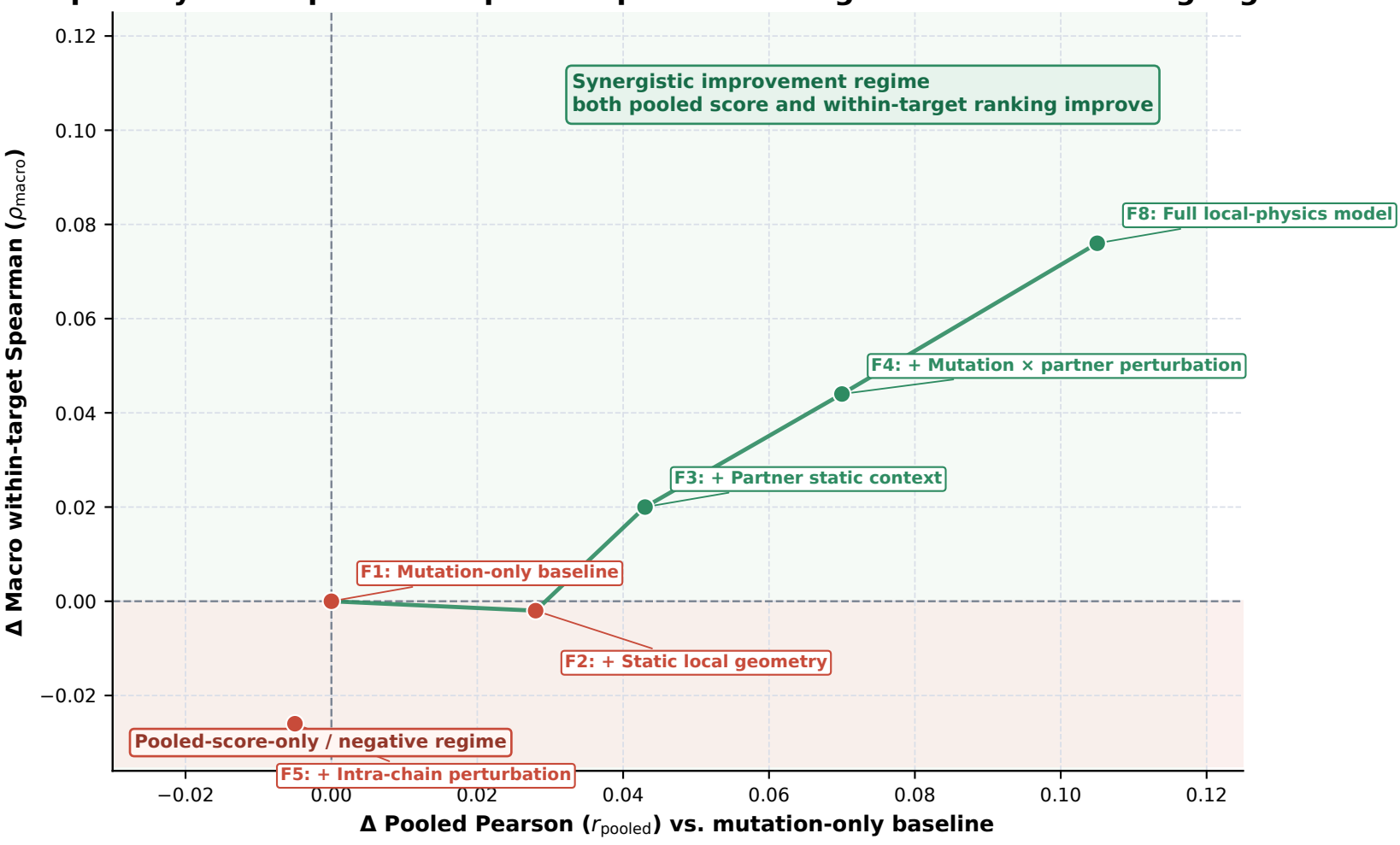
Capability decomposition separates pooled-score gains from within-target gains
Synergistic improvement regime
both pooled score and within-target ranking improve
F8: Full local-physics model
F4: + Mutation × partner perturbation
F3: + Partner static context
F1: Mutation-only baseline
F2: + Static local geometry
Pooled-score-only / negative regime
F5: + Intra-chain perturbation
Δ Macro within-target Spearman (ρ_{macro})
Δ Pooled Pearson (r_{pooled}) vs. mutation-only baseline
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.02
0.00
0.02
0.04
0.06
0.08
0.10
0.12